MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY TTL,
SHIFT REGISTERS, CASCADABLE, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/306D, dated
6 November 1987, and is approved for use by all
Departments and Agencies of the Department of Defense.

PAGE 1

Title: Delete and substitute as printed above.

PAGE 6

TABLE I, f_{MAX}, device type 09, minimum limits column: Delete "22" and substitute "20".

PAGE 8

TABLE I, test column: Delete "Output disable time to low level" and substitute "Output disable time from low level".

TABLE I, test column: Delete "Output disable time to high level" and substitute "Output disable time from high level".

PAGE 9

4.2a(1), delete and substitute as follows:

"(1) Test condition D or E, using the circuit shown on figure 13, or equivalent. Test condition A and the applicable test circuit shall be allowed with the approval of the qualifying activity."

PAGE 10

4.4.3b(1), delete and substitute as follows:

"(1) Test condition D or E, using the circuit shown on figure 13, or equivalent. Test condition A and the applicable test circuit shall be allowed with the approval of the qualifying activity."

PAGE 11

FIGURE 1, device type 03, cases outlines A, B, C, and D, pin 9: Delete "CLK/" and substitute "CLK1". Cases outlines 2 and X, pin 13: Delete "CLK/" and substitute "CLK1".

PAGE 13

FIGURE 1, device type 09, cases outlines E and F, pin 6: Delete "CLOCK" and substitute "CLOCK INHIBIT".

PAGE 16

FIGURE 2, device type 03, notes, third line: Delete " $\uparrow$ = Transition from high to low level" and substitute " $\downarrow$ = Transition from high to low level".

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PAGE 24

FIGURE 3, Logic diagram, device type 03, circuit A: Delete in its entirety.

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FIGURE 3, Logic diagram, device type 03: Delete "CIRCUIT B, C, D, E" and substitute "CIRCUITS A, B, C, D, E".

PAGE 49

FIGURE 10, device type 07, after OPTIONAL OUTPUT CIRCUIT: Add "(ALL OUTPUTS)".

PAGE 67

TABLE III, I_{IL2}, I_{IL3}, I_{IL4}, I_{OS}, AND I_{CC}, test numbers 46 through 58, V_{CC} column: Add "„".

TABLE III, I_{IL4}, test number 53, CLR column: Add "4.5 V".

TABLE III, I_{CC}, test number 58, minimum limits column: Delete ",,".

PAGE 68

TABLE III, truth table test, test number 68, S/L serial column: Delete "C" and substitute "B".

PAGE 70

TABLE III, footnote 2/: Delete and substitute as follows:

"2/ I_{IL} Limits (mA) min/max values for circuits shown:

Parameter	Terminal	A	B	C	D	E	F	G
I_{IL1}	CLR	-.16/-.4	-.11/-.35	-.16/-.4	-.12/-.35	-.12/-.36	-.12/-.36	-.16/-.4
I_{IL2}	S/R, A_{IN}, B_{IN}, C_{IN}, D_{IN}, S/L	"	-.11/-.35	"	-.16/-.4	-.105/-.345	"	"
I_{IL3}	S_0, S_1	"	-.03/-.3	"	-.12/-.36	-.12/-.36	"	"
I_{IL4}	CLK	"	-.03/-.3	-.20/-.44	-.12/-.36	-.12/-.36	"	-.15/-.38

PAGE 75

TABLE III, footnote G, first line: Delete "f_{MAX} maximum" and substitute "f_{MAX} minimum". Fourth line: Delete "input ↑ and ↑ " and substitute "input ↑ and ↓ ".

PAGE 77

TABLE III, I_{CC}, test number 47, minimum limits column: Delete "„"

PAGE 78

TABLE III, f_{MAX}, test number 84, CLK_2 column: Delete "„".

TABLE III, t_{PHL1}, test numbers 93 through 100, minimum limits column: Add "„".

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TABLE III, truth table test, test number 59, enable column: Delete "„" and substitute "B".

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TABLE III, t_{PHL1} and t_{PHL2}, test numbers 97 through 106, GND column: Add "„".

PAGE 83

TABLE III, note E, third line: Delete "the serial ↑ and ↑ "and substitute "the serial ↑ and ↓ ".

PAGE 84

TABLE III, I_{IL1}, test numbers 29 through 32, V_{CC} column: Add "„".

PAGE 87

TABLE III, footnote J: Delete and substitute as follows:

"J. Apply input pulse: ___/ _____ 2.5 V minimum/5.5 V maximum
0 V "

PAGE 92

TABLE III, footnote 1/: Delete and substitute as follows:

"1/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Terminal	A	B	C	D	E
I_{IL1}	Serial	-.075/-.250	-.16/-.40	-.16/-.40	-.105/-.345	-.12/-.36
	A_{IN}, B_{IN}, C_{IN}, D_{IN}	-.12/-.36	-.16/-.40	-.16/-.40	-.105/-.345	-.12/-.36
	Mode	-.16/-.40	-.15/-.38	-.03/-.3	-.12/-.36	-.12/-.36
	CONT	-.16/-.40	-.16/-.40	-.03/-.3	-.12/-.36	-.12/-.36
	CLK	-.16/-.40	-.20/-.44	-.03/-.3	-.12/-.36	-.12/-.36

"

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TABLE III, I_{OS}, test numbers 55 through 59, V_{CC} column: Add "„".
TABLE III, I_{OS}, test numbers 55 through 59, Measured terminal column: Delete and substitute:

" Q_A
Q_B
Q_C
Q_D
$Q_{D'}$ "

PAGE 95

TABLE III, t_{PLH1}, test numbers 96 through 98, serial column: Add "„".

TABLE III, t_{PHL1}, test numbers 106 through 108, serial column: Add "„".

PAGE 97

Footnote 2/, Delete and substitute:

" 2/ I_{OS} limits for circuit A for QA through QD are -30 to -130 mA, for QD' is -20 to -100 mA, and for circuits B, C, and D are -15 to -100 mA.

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TABLE III, I_{IL6}, test numbers 24 through 26, shift load column: Delete "„".

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TABLE III, truth table test, subgroup column: Delete "1/" and substitute "2/".

TABLE III, truth table test, symbol column: Delete "f_{MAX} 3/" and substitute " f_{MAX} 4/".

TABLE III, truth table test, min test limits column: Delete "2/" and substitute "3/".

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TABLE III, the first footnote 1/: Delete "1/ A = 2.5 V and B = 0.4 V" and substitute "2/ A = 2.5 V and B = 0.4 V".

TABLE III, footnote 2/: Delete " 2/ Ouput voltage ... " and substitute " 3/ Output voltage ... ".

TABLE III, footnote 3/: Delete "3/ f_{MAX}..." and substitute "4/ f_{MAX}..."

TABLE III, delete the second footnote " 1/ I_{IL}..." and substitute as follows:

" 1/ I_{IL} limits (mA) min/max values for circuit shown:

Parameter	Terminal	A	C	F
I_{IL1}	CLK, CLK/INHIB	-.001/-.150	-.12/-.38	-.005/-.72
I_{IL6}	A, B, C, D, E, F, G, H	-.120/-.360	-.12/-.38	-.12/-.38
	S/IN	-.100/-.340	-.12/-.38	-.12/-.38
I_{IL7}	S/L	-.001/-.150	-.36/-1.08	-.005/-.72

"

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TABLE III, subgroup 10, f_{MAX}, minimum limits column: Delete "22" and substitute "20".

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI, SM
Navy - OS, SH, TD, AS, CG, MC
Air Force - 19, 85, 99
DLA - ES

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-E093-97)

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